

Math proofs illustrated by sequence diagrams

This document pairs short mathematical derivations with mermaid sequence diagrams that narrate, step by step, what the proof is *doing*. The math renders through KaTeX in HTML mode; the diagrams render through mermaid.js in the same `foreignObject` channel. Both engines coexist in the same flow.

1. The sum of the first n integers

The classical Gauss anecdote: pair the first and last terms of the sum $1 + 2 + \dots + n$, then the second and second-last, and so on. Each pair sums to $n+1$, and there are $n/2$ pairs.

$$S = 1 + 2 + \dots + n \quad 2S = (n+1) + (n+1) + \dots + (n+1) = n(n+1) \quad S = \frac{n(n+1)}{2}.$$

The pairing as a sequence of moves, with a "scribe" actor that writes the intermediate result and a "checker" that verifies it:

[Mermaid diagram omitted in non-SVG back-end]

2. The infinite geometric series

For $|r| < 1$, the geometric series $\sum_{k=0}^{\infty} r^k$ converges to $1/(1-r)$. The proof multiplies the partial sum S_n by r , subtracts, and takes the limit.

$$\begin{aligned} S_n &= 1 + r + r^2 + \dots + r^n \\ rS_n &= r + r^2 + \dots + r^{n+1} \\ S_n - rS_n &= 1 - r^{n+1} \\ (1-r)S_n &= 1 - r^{n+1} \\ S_n &= \frac{1 - r^{n+1}}{1-r} \end{aligned} \quad \xrightarrow{n \rightarrow \infty} \frac{1}{1-r}.$$

As a dialogue between the "student" who proposes each manipulation and the "book" that confirms what is allowed:

[Mermaid diagram omitted in non-SVG back-end]

3. The derivative of x squared from first principles

The limit definition of the derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

For $f(x) = x^2$:

$$\frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \xrightarrow{h \rightarrow 0} 2x.$$

The same calculation as a back-and-forth between "calculus" and "algebra":

[Mermaid diagram omitted in non-SVG back-end]

4. The Cauchy-Schwarz inequality

For real vectors $\mathbf{u}, \mathbf{v}$, the inequality $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ follows from the non-negativity of the squared norm $\|\mathbf{u} - t\mathbf{v}\|^2$ for all real t . The discriminant of the resulting quadratic in t must be non-positive.

$$\|\mathbf{u} - t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2 \geq 0.$$

Treating this as a quadratic in t with non-positive discriminant:

$$\frac{4\langle \mathbf{u}, \mathbf{v} \rangle^2 - 4\|\mathbf{u}\|^2 \|\mathbf{v}\|^2}{4} \leq 0 \implies \langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2.$$

The proof as a passing-the-baton between three reviewers:

[Mermaid diagram omitted in non-SVG back-end]

Section summary

Each of the four proofs exercises a single algebraic move (pairing, telescoping, the limit definition, a non-negative quadratic) and each is shadowed by a mermaid sequence diagram in which the move is voiced by named actors. The HTML build engraves all four diagrams alongside the KaTeX-rendered math; the PDF build keeps the math via `@Math`'s placeholder and renders each mermaid block as [Mermaid diagram omitted in non-SVG back-end]. For an archival PDF that includes the diagrams, pre-render each fence with `mmdc -i diag.mmd -o diag.svg` and inline the result via ``.